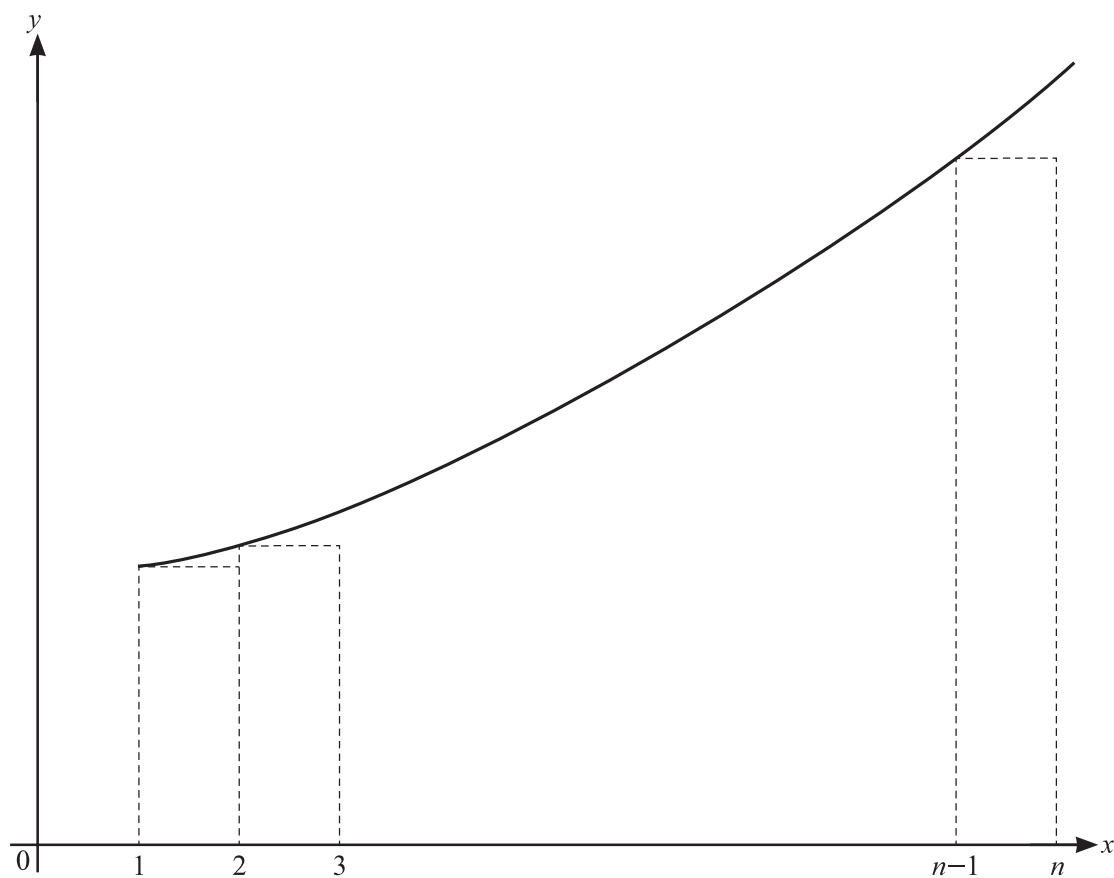


4



The diagram shows the curve with equation  $y = \frac{1}{\sqrt{x}}e^{\sqrt{x}}$  for  $x \geq 1$ , together with a set of  $n-1$  rectangles of unit width.

(a) By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^n \frac{1}{\sqrt{r}} e^{\sqrt{r}} < \left(2 + \frac{1}{\sqrt{n}}\right) e^{\sqrt{n}} - 2e. \quad [5]$$

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**[Turn over**